

EU SAE Application Package

Download and verification instructions

5.2.0-rc.6-wizard.3-pointwise

Release-candidate warning. Public sharing remains blocked until Git history/release assets and institutional rights/authority are remediated. Automatic source archives for superseded releases must not be used. A public URL is intentionally not supplied here.

1. Obtain the correct archive

Obtain the curated, history-free wizard archive from the authorized release manager. Its top-level folder must be `EU_SAE_wizard_5.2.0-rc.6-wizard.3-pointwise`. Do not zip the source workspace and do not use an automatic repository source archive.

2. Verify the package

Open `VERSION` and confirm `5.2.0-rc.6`, and open `WIZARD_VERSION` and confirm `5.2.0-rc.6-wizard.3-pointwise`. Then verify every listed file against `release_manifest.csv` using SHA-256. Stop if a listed file is missing or changed, or if an unexpected file is present. Use `Rscript scripts/verify_release.R` from the extracted package root to check the full file set and hashes; compare the ZIP hash with `SHA256SUMS.txt`.

3. Check the expected contents

The clean package contains the application, R helpers, scripts, tests, output placeholders, `docs/instructions/`, `docs/guidance/`, licence and notice files, and both example datasets under `Data/simulated/` and `Data/Spain/`. The literature folder `docs/guidance/literature/` contains only `README.md`; other literature files are excluded. Internal notes, `.git`, `app_runs`, user-created Data folders, local libraries, credentials, generated outputs and local diagnostics must not ship. The Spain survey and auxiliary files carry their documented GPL-2 terms. The boundary file uses documented IGN/CNIG CartoBase ANE under CC BY 4.0; see `Data/Spain/README.md` and `THIRD_PARTY_NOTICES.md`. The Spain survey derivation script remains missing. This review candidate is not approved for public distribution.

4. Install and start

Install R 4.2 or later and run `install_packages.R` once. This package provides two interfaces over the same analysis engine and the same control set; they may run at the same time. On Windows, open `Start_Here: Start_Wizard.bat` opens the guided wizard on port 7788; `Start_Dashboard.bat` (or `bash Start_Dashboard.sh` from the package root on macOS/Linux) opens the classic dashboard on port 7777. Both are covered by the slide deck in `docs/instructions/`; `docs/README_WIZARD.md` holds the wizard reference. Keep the launcher window open while using the local application. HTML and Word reports require Pandoc (standalone or bundled with RStudio); `install_packages.R` supplies `xml2` and `zip` for Word formatting.

5. Follow the wizard flow (or the dashboard equivalent)

The wizard presents the same inputs as six sequential steps: 1 Data (country, analysis years, seed, run label, input files), 2 Mapping (survey columns and join keys), 3 Indicator (FGT or mean welfare), 4 Models (UFH/MFH options, MCPE replicates, benchmarking, covariate selection, PSU consistency), 5 AI Assistant (optional consent, key, language) and 6 Review & Run (settings summary, then 1. Check Data Readiness followed by 2. Run Analysis, with the Preflight, Data Readiness and Status panels beneath). The breadcrumb is clickable: green means complete, amber means an outstanding item, blue means the current step. Navigation is soft - Next stays available on an incomplete step and the outstanding items are listed - but the run control stays disabled until the fatal data prerequisites are present. Load Last Setup is available from every step; API keys and transfer consent are never persisted. In the classic dashboard the same controls sit in one sidebar with the same two run buttons.

Input data, model options, reproducibility and optional AI

6. Select approved local inputs

To learn the workflow, use either bundled example. Data/simulated/ holds fully synthetic files - a survey, auxiliary covariates, and geometry, with a suggested variable mapping in that folder's README and the seeded generator script that produced them; they contain no real data of any kind. Data/Spain/ holds the Spain province example used in the guidance notes, with its mapping, attribution and licence in Data/Spain/README.md. Neither example may ever feed a real estimate. For actual analyses, browse to approved household survey microdata, domain-level auxiliary covariates, geometry, and any optional population or benchmark-target files, confirming provenance, permitted use, sensitivity classification, area codes, and year coverage before running. You may group your approved inputs in a folder such as Data/Romania_2025/; Browse also accepts files elsewhere. Results remain under outputs/. See Data/README.md for saved-setup compatibility.

7. Apply the revised variance rule

For both UFH and MFH, sm_out uses one threshold rule: replace the direct sampling variance with its smoothed variance only when the direct value is missing/non-finite or strictly below 0.001. A direct variance equal to 0.001 is retained. sm_all replaces all direct variances; direct retains the direct values subject to the documented safety backfills.

8. Choose the benchmarking level explicitly

After selecting Apply benchmarking, choose National or Grouped under Benchmarking level. National benchmarking uses one countrywide group and constrains the population-weighted domain estimates to the direct national target. Grouped benchmarking requires a mapped higher-level survey variable, such as region or NUTS2, and applies the constraint separately within each group. An external Benchmark Target Database remains optional.

9. Record reproducibility and inference controls

The analysis seed defaults to 123. The MFH MCPE bootstrap uses mcpe_nB (interactive default 200; minimum 50); use at least 500 replications for production inference and assess Monte Carlo stability. The principal significance flag and red/gray figures use the pointwise raw test. At $\alpha = 0.05$, significance matches whether the pointwise 95% confidence interval excludes zero. Benjamini-Hochberg fields remain supplementary sensitivity outputs: they control the expected share of false flags among flagged domains and should be used for any statement about how many domains changed.

10. Understand the model selection rule

MFH2 is the default MFH model in both the wizard and the classic dashboard. Selecting MFH3 invokes the Molina-Romero sequence: MFH3 is fitted first; an estimation error or a returned non-converged fit moves automatically to MFH2, and the two cases are labeled separately in the run log. A converged MFH3 fit is retained only when its random-effect variance-homogeneity test supports heterogeneity after the selected multiplicity adjustment (Bonferroni by default; raw, Bonferroni and BH results are all exported). MFH1 remains available as a manually selected sensitivity model. The optional MFH3 sensitivity fit does not change the selected model. Section 2 Run Setup of the final report records which model was used.

11. Interpret covariates and fallbacks correctly

With LASSO enabled, year-specific UFH or MFH covariate lists define the screening candidate pool. With LASSO disabled, they define the fixed specification. If a boundary MFH fit cannot be benchmarked, Comparison records the reason in benchmark_status.csv and continues with unbenchmarked estimates.

12. Treat AI as a separate external transfer

AI is off by default. Enter a provider key only after selecting the separate external-transfer acknowledgment and confirming that the provider or configured gateway is institutionally approved. The application does not send household microdata. Prompts carry unpublished aggregate estimates, uncertainty measures, diagnostics, counts and analysis context; geographic identifiers are removed from the top-ranked and extreme-change prompt tables. Aggregate unpublished estimates may still be confidential in context, so authorization is still required. AI commentary is interleaved with the corresponding statistical sections in final_report.html and final_report.docx, is advisory, and requires documented human review before dissemination.

Recordkeeping and release gate

13. Read both reports and keep the evidence

After a successful run, open `outputs/final_report.html` in a browser or `outputs/final_report.docx` in Word. Word text and tables are editable, and figures are embedded images. Save a separate working copy before editing; later runs replace current outputs. Both formats include the same statistical results and labeled AI text when requested. Keep `outputs/data/` alongside the reports for linked Excel workbooks; archived copies are under `app_runs/<timestamp>_<run_label>/outputs/`. The clean ZIP excludes generated reports. The new signed-change boxplot and paired scatterplot are under `outputs/figures/change_comparisons/`; source values are in `outputs/data/change_estimate_comparison.xlsx` and `EU_SAE_results.xlsx`. These compare estimated changes in percentage points, not confidence-interval widths. The original CI-width figures remain. A tighter spread across domain estimates alone does not establish smaller MSE or a statistically significant variance reduction. If Word conversion fails, the HTML remains; check the run log, Pandoc, xml2 and zip, and regenerate after resolving the warning. Retain the exact archive, `release_manifest.csv`, verification output, `VERSION`, `WIZARD_VERSION`, run metadata, input hashes, and review approvals. Passing tests or checksums does not approve the statistical methods, data rights, privacy basis, or official publication. Release gate. Before any public release, complete `docs/HISTORY_REMEDIATION.md` and `docs/RELEASE_CHECKLIST.md`, obtain institutional approval for documentation/media rights and release authority, re-clone the rewritten repository, rebuild the clean archive, and independently verify its manifest.